

Introduction to Microeconomics

Supply & Demand

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Years
Anniversary

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- Why do male soccer players earn more than female soccer players?
- Why do diamonds cost more than water?
- Why do Picasso's paintings sell for more than other artists?
- Is it the cost of production that determines prices, or is it willingness to pay that determines prices?
- Alfred Marshall, in *Principles of Economics*, 1890 was the first to explain clearly how both costs and willingness to pay interact to determine market prices.

Definition

The **market** for any good or service consists of all (actual or potential) buyers or sellers of that good or service.

The demand curve

The **demand curve** for a good or service tells us the total quantity of that good or service that buyers wish to buy at each price.

Graphical representation of the demand curve

Movements along the demand curve and shifts of the demand schedule

Producers: the supply curve

The supply curve

The **supply curve** of a good or service tells us the total quantity of that good or service that sellers wish to sell at each price.

Graphical representation of the supply curve

Movements along the supply curve and shifts of the supply schedule

Consumer Surplus

Consumer Surplus is the excess of the benefit received from a good over the amount paid for it. We can calculate consumer surplus as the marginal benefit (or value) of a good minus its price, summed over the quantity bought. It is measured by the area under the demand curve and above the price paid, up to the quantity bought.

Producer Surplus

Producer Surplus is the excess of the amount received from the sale of a good over the cost of producing it. We calculate it as the price received for a good minus the minimum-supply price (marginal cost), summed over the quantity sold. On a graph, producer surplus is shown by the area below price received and above the supply curve, summed over the quantity sold.

Market Equilibrium price and Quantity

Equilibrium occurs at the price-quantity pair for which both buyers and sellers are satisfied.

Excess Supply and Excess Demand

Movement towards market equilibrium

Social Surplus

Social surplus is the sum of producer and consumer surplus. It represents the total gains from participating in a market.

The Equilibrium Principle

A market in equilibrium leaves no unexploited opportunities for individuals, but may not exploit all gains achievable through collective action.

Linear demand and supply curves

Equilibrium with linear supply and demand curves

Comparative Statics Analysis (Ceteris Paribus Analysis)

Aggregate Demand and Supply

Suppose that individual demand curves are given by $q_i^d(p)$, for $i = 1, \dots, I$ consumers; and individual supply curves are given by $q_k^s(p)$, for $k = 1, \dots, K$ producers/firms. The **aggregate or market demand** is obtained by **horizontal summation** of all the individual demand curves— at price p , we add up all demanded quantities:

$$Q^D(p) = \sum_{i=1}^I q_i^d(p).$$

Similarly, the **aggregate or market supply** is obtained by **horizontal summation** of all the individual supply curves—at price p , we add up all supplied quantities:

$$Q^S(p) = \sum_{k=1}^K q_k^s(p).$$

Warning!

When doing the horizontal summation of the individual demand and supply curves to obtain the aggregate curves, you must always be careful to add them up at prices that lead to **positive quantities!** Remember, individual demand and supply curves are only defined for positive prices and positive quantities!

